# 1 Introduction

Term deposits are a major source of income for a bank. A term deposit is a cash investment held at a financial institution. Your money is invested for an agreed rate of interest over a fixed amount of time, or term. The bank has various outreach plans to sell term deposits to their customers such as email marketing, advertisements, telephonic marketing, and digital marketing.

Telephonic marketing campaigns still remain one of the most effective way to reach out to people. However, they require huge investment as large call centers are hired to actually execute these campaigns. Hence, it is crucial to identify the customers most likely to convert beforehand so that they can be specifically targeted via call.

In the competitive landscape of the banking sector, understanding customer behavior is paramount for optimizing marketing strategies and enhancing operational efficiency. One critical area of interest is predicting whether clients will subscribe to term deposits, a financial product that offers banks stable funding and clients attractive returns. Accurate prediction of term deposit subscriptions enables banks to target potential customers effectively, reduce campaign costs, and improve resource allocation. However, the complexity of customer decision-making, influenced by demographic, financial, and behavioral factors, necessitates advanced analytical approaches.

This study leverages a comprehensive dataset from a banking institution, encompassing attributes such as age, occupation, marital status, education level, account balance, housing and loan status, contact method, and campaign-related details (e.g., contact duration, number of contacts, and previous campaign outcomes). The dataset, comprising thousands of client records, provides a rich foundation for modeling the binary outcome variable, indicating whether a client subscribed to a term deposit. The primary objective is to develop and compare predictive models using machine learning techniques, including logistic regression, decision trees, random forests, and support vector machines (SVM), to accurately classify subscription outcomes.

Machine learning has emerged as a powerful tool in financial analytics, offering the ability to uncover patterns and relationships in large, heterogeneous datasets. By applying these methods, this study aims to identify key predictors of term deposit subscriptions and evaluate the performance of each algorithm using standard metrics such as accuracy, precision, recall, and F1-score. The findings are expected to provide actionable insights for banking institutions, enabling data-driven marketing strategies and contributing to the growing body of research on predictive modeling in finance.

# 2 Dataset Description

The dataset used in this study is sourced from Kaggle, originating from direct marketing campaigns conducted via phone calls by a Portuguese banking institution. It is designed to support the binary classification task of predicting whether a client will subscribe to a term deposit, represented by the target variable `y` (yes or no). The dataset comprises 45,211 records, each characterized by 17 attributes that capture demographic, financial, and campaign-related information about clients. These variables provide a comprehensive basis for modeling client behavior in response to telephonic marketing efforts.

The attributes in the dataset are as follows:

- `age`: Numeric, representing the client's age in years.
- `job`: Categorical, indicating the client's occupation (e.g., management, technician, blue-collar, retired).

- `marital`: Categorical, denoting marital status (married, single, divorced).
- `education`: Categorical, specifying the highest education level attained (primary, secondary, tertiary, unknown).
- `default`: Binary, indicating whether the client has credit in default (yes or no).
- `balance`: Numeric, representing the average yearly account balance in euros.
- `housing`: Binary, indicating whether the client has a housing loan (yes or no).
- `loan`: Binary, indicating whether the client has a personal loan (yes or no).
- `contact`: Categorical, specifying the communication method (cellular, telephone, unknown).
- `day`: Numeric, representing the day of the month when the client was last contacted.
- `month`: Categorical, indicating the month of the last contact (e.g., jan, feb, ..., dec).
- `duration`: Numeric, representing the duration of the last contact in seconds.
- `campaign`: Numeric, indicating the number of contacts performed during the current campaign for the client.
- `pdays`: Numeric, representing the number of days since the client was last contacted in a previous campaign (-1 indicates no prior contact).
- `previous`: Numeric, indicating the number of contacts performed before the current campaign for the client.
- `poutcome`: Categorical, representing the outcome of the previous marketing campaign (success, failure, other, unknown).
- `y`: Binary, the target variable indicating whether the client subscribed to a term deposit (yes or no).

The dataset includes a mix of numeric and categorical variables, with some attributes (e.g., `poutcome`, `contact`) containing 'unknown' values that require preprocessing. The target variable `y` is imbalanced, with a smaller proportion of positive (yes) outcomes, posing a challenge for classification models. This rich dataset enables the application of machine learning techniques, such as logistic regression, decision trees, random forests, and support vector machines, to uncover patterns predictive of term deposit subscriptions.

The training dataset (`train.csv`) contains 45,211 rows and 18 columns, ordered by date from May 2008 to November 2010. .

# 3 Objectives

The objectives of this study are:

- To check whether there are any missing values in the dataset.
- To show basic plots for data exploration.
- To perform classification using Decision Tree, Random Forest, SVM and Xgboost.
- To evaluate model performance using accuracy, F1-score, precision, and other relevant metrics.

# 4 Check Missing Values

To ensure the integrity of the Kaggle Bank Marketing dataset, comprising `train.csv` (45,211 rows, 18 columns) and `test.csv` (4,521 rows, 18 columns), a thorough examination for missing values was conducted across all columns. This step is critical as missing data can bias model predictions and reduce the effectiveness of classification algorithms like Decision Trees, Random Forests, and Logistic Regression. The dataset includes 18 variables: `age`, `job`, `marital`, `education`, `default`, `balance`, `housing`, `loan`, `contact`, `day`, `month`, `duration`, `campaign`, `pdays`, `previous`, `poutcome`, `ID`, and `y`.

The analysis revealed no explicit missing values (e.g., `NULL` or `NaN`) in any column of either dataset. Additionally, categorical variables such as `education`, `contact`, and `poutcome` were inspected for implicit missing values represented as 'unknown'. In this dataset, 'unknown' entries are treated as a valid category rather than missing data, as they provide meaningful information about the absence of specific knowledge (e.g., unknown education level or contact method). Thus, no imputation was required.

Had missing values been present, the following imputation strategies would have been applied:

- **Categorical Variables**: For variables like `job`, `marital`, `education`, `default`, `housing`, `loan`, `contact`, `month`, and `poutcome`, missing values would be imputed with the mode (most frequent category). This approach preserves the distribution of categorical data and is computationally efficient.
- **Numeric Variables**: For variables like `age`, `balance`, `day`, `duration`, `campaign`, `pdays`, and `previous`, missing values would be imputed using one of the following methods based on data characteristics:
  - **Mean or Median**: The mean is used for normally distributed variables, while the median is preferred for skewed distributions (e.g., `balance`, `duration`) to reduce the impact of outliers.
  - **K-Nearest Neighbors (KNN)**:Fill missing values using average values from nearest neighbors.
  - **Regression-Based Prediction**: A regression model trained on other features would predict missing values, capturing underlying patterns but increasing computational complexity.

The choice of imputation method would depend on the extent of missing data and variable distribution, assessed via histograms and statistical summaries. Since no missing values were found, the dataset proceeded to subsequent preprocessing steps without imputation, ensuring robust data quality for exploratory analysis and classification tasks.
There is also no duplicate value present in the dataset.

# 5 Outlier Detection

Outlier detection was performed on the Kaggle Bank Marketing dataset (`Term deposit train data.csv`, 45,211 rows, 17 columns) to identify extreme values in numeric variables (`age`, `balance`, `duration`, `campaign`, `pdays`, `previous`) that could affect the performance of classification models (Decision Trees, Random Forests, Logistic Regression). The variable `day` (day of the month, 1–31) was excluded due to its bounded nature, which makes outliers unlikely. Two methods were used: the Interquartile Range (IQR) method and the Z-score method, implemented in R. Below, we describe the methodology, findings, and treatment plan.

## 5.1 Methodology

- **Visual Method**:Boxplot ,Histogram is the visual method to check outliers.
- **IQR Method**: For each variable, the first quartile ($Q1$), third quartile ($Q3$), and interquartile range ($\text{IQR} = Q3 - Q1$) were calculated. Values below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$ were flagged as outliers. This method is robust to non-normal distributions, common in variables like `balance` and `duration`.
- **Z-score Method**: The Z-score, $Z = \frac{x - \mu}{\sigma}$, measures how many standard deviations a value is from the mean. Values with $|Z| > 3$ were flagged as outliers. This method assumes approximate normality, which may limit its effectiveness for skewed variables.

- **Special Consideration for `pdays` and `previous`**: The variable `pdays` has many -1 values (indicating no prior contact), which are not outliers. Positive `pdays` values (indicating prior contact) were analyzed for outliers. Similarly, `previous` has many zeros, and non-zero values were assessed.

## 5.2 Findings

The outlier counts and example indices are summarized below, based on the training dataset:

- **Visual Method**:

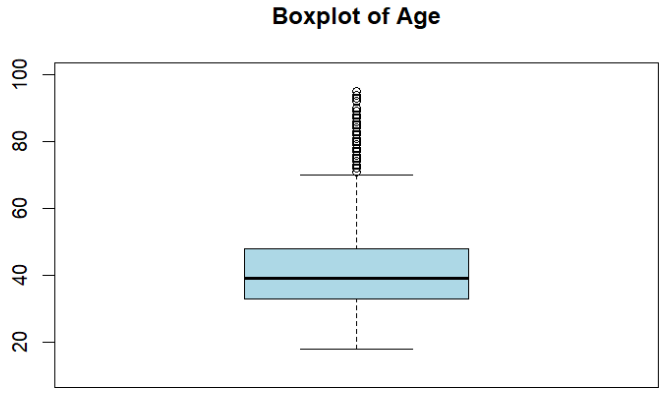
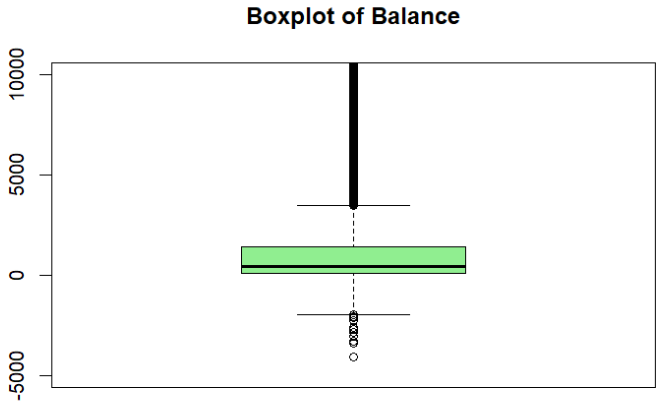


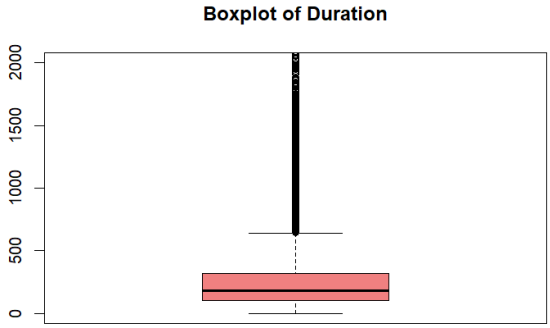

Figure 1: Boxplot for Age

- **age** (Min: 18, Median: 39, Max: 95):
  - **IQR**: 487 outliers (e.g., ages 89, 81, 82), values $> 70$.
  - **Z-score**: 381 outliers (e.g., ages $> 72$.
  - **Note**: Few outliers reflect `age`'s near-normal distribution.
- **balance** (Min: -8019, Median: 448, Max:102127):
  - **Visual Method**:




  - **IQR**: 4729 outliers,There have some negative and positive sign outliers
  - **Z-score**: 745 outliers, values $> 5768$ or $< -172$.
  - **Note**: High outlier count due to right-skewed distribution (e.g., large deposits).

- **duration** (Min: 0, Median: 180, Max: 4918):
  - **Visual Method**:




  - **IQR**:3235 outliers values $> 643$.
  - **Z-score**: 963 outliers , values $> 1030$.
  - **Note**: Skewed distribution leads to many long call durations flagged.

- **campaign** (Min: 1, Median: 2, Max: 63):
  - **IQR**: 3064 outliers , values $> 7$.
  - **Z-score**: 840 outliers , values $> 12$.
  - **Note**: Skewed toward low contact counts; high values are rare.

- **pdays** (Min: -1, Median: -1, Max: 871):
  - **IQR**: 8257 outliers, any value $> 0$.
  - **Z-score**: 1723 outliers, values $> 340$.
  - **Note**: Positive `pdays` (prior contact) are not inherently outliers; only extreme values are considered.

- **previous** (Min: 0, Median: 0, Max: 275):
  - **IQR**: 8257 outliers , any value $> 0$.
  - **Z-score**: 582 outliers , values $> 8$.
  - **Note**: Non-zero values align with positive `pdays`; only extreme values are outliers.

## 5.3 Explanation and Treatment Plan

The IQR method flagged more outliers than the Z-score method for `balance`, `duration`, `campaign`, `pdays`, and `previous` due to their skewed distributions, where IQR's non-parametric approach is more robust. The Z-score method, assuming normality, was more conservative and effective for `age`. For `pdays` and `previous`, positive values reflect prior campaign interactions and were only treated as outliers if extremely high (e.g., `pdays` $> 340$).

**Treatment Strategy**:

- **age**: Outliers are retained, as they are plausible (elderly clients) and few in number.
- **balance**: Outliers are capped at the 5th and 95th percentiles to reduce skewness while preserving data.
- **duration**: Outliers are capped at the 95th percentile to mitigate the impact of long calls.
- **campaign**: Outliers are capped at the 95th percentile to handle rare high-contact cases.

- **pdays** and `previous`: Positive values are retained unless extreme (e.g., `pdays` $> 340$, capped at 340; `previous` $> 8$, capped at 8). The -1 in `pdays` and 0 in `previous` are kept as valid categories.

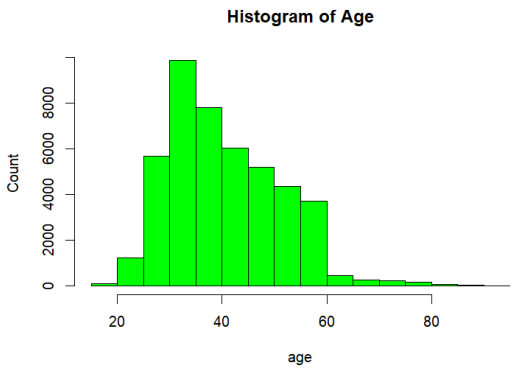
No rows were removed to preserve the dataset's size (45,211 rows). Capping ensures robust input for subsequent classification tasks while maintaining data integrity. The treated dataset will be used for exploratory plots and model training.

# 6 Basic Plot Of the Data(e.g-Histogram and Barplot)

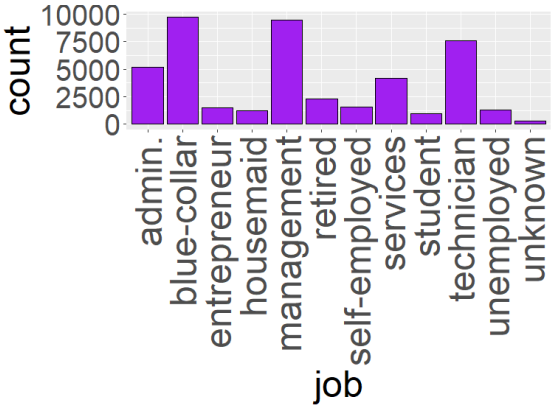
After capping the extreme values of the `balance` variable, some negative values remained in the dataset. Since the presence of negative values makes analysis difficult, we applied a transformation to handle this issue. Specifically, we used the following formula:

$$\log 1p\,(\texttt{data} + |\min(\texttt{data})| + 1)$$

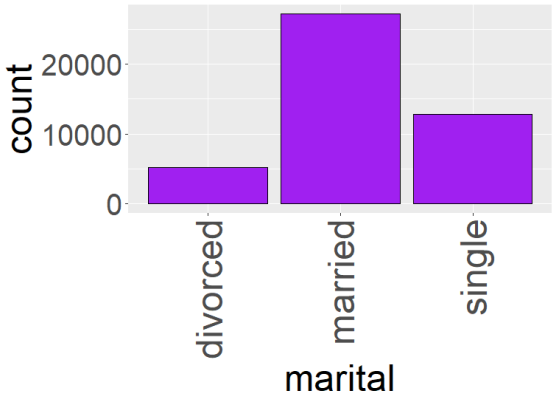
This transformation ensures that all values become positive before applying the logarithm, thereby stabilizing the variance and making the data more suitable for analysis.



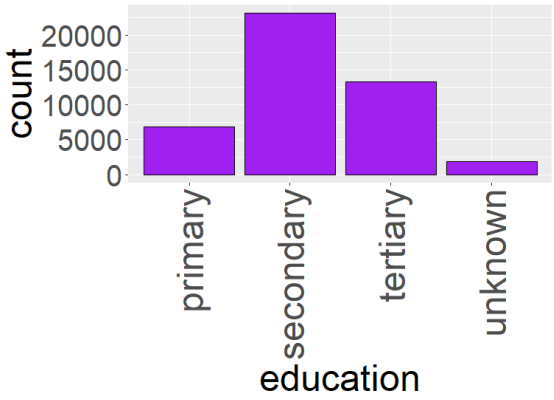

(a) Image 1



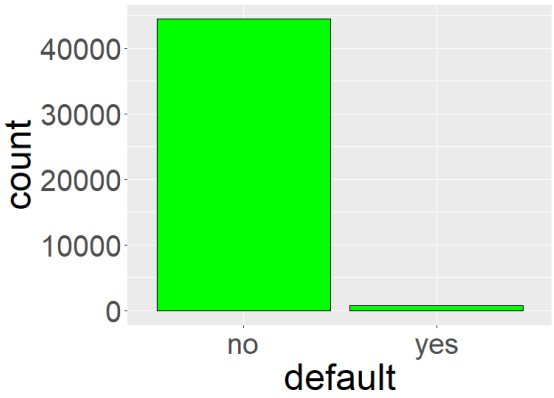

(b) Image 2



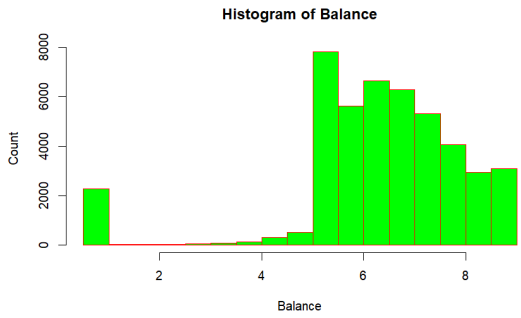

(c) Image 3



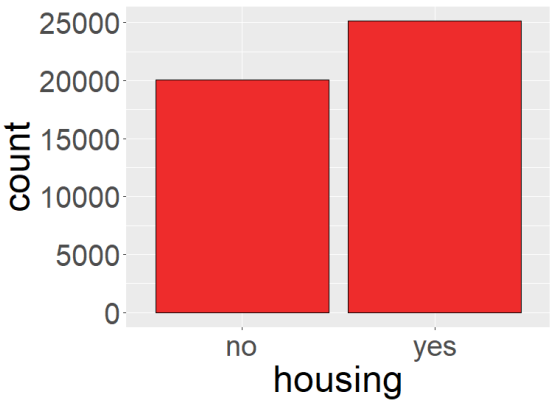

(d) Image 4



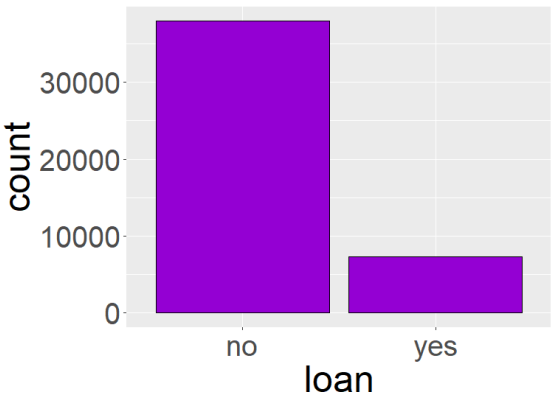

(e) Image 5



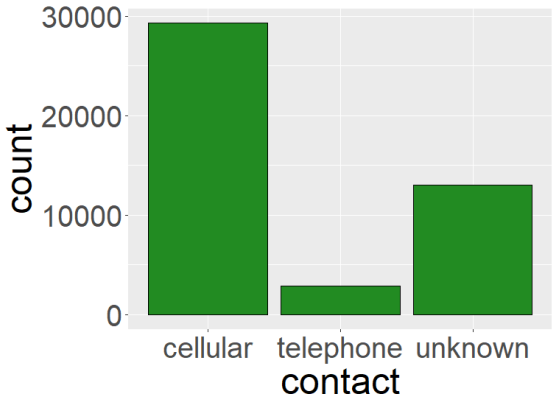

(f) Image 6



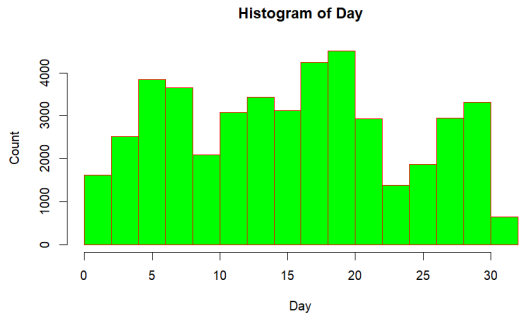

(g) Image 7



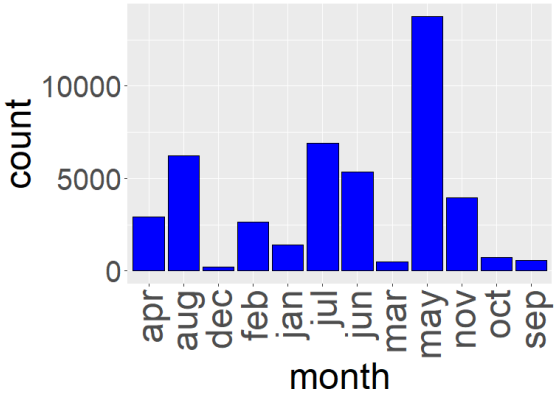

(h) Image 8



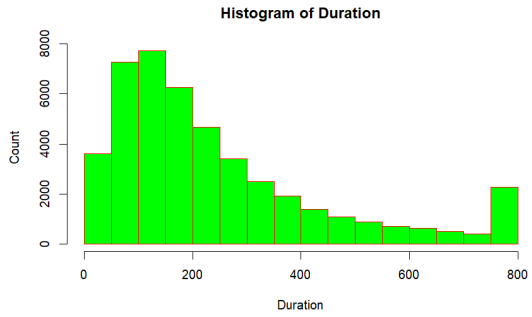

(a) Image 9



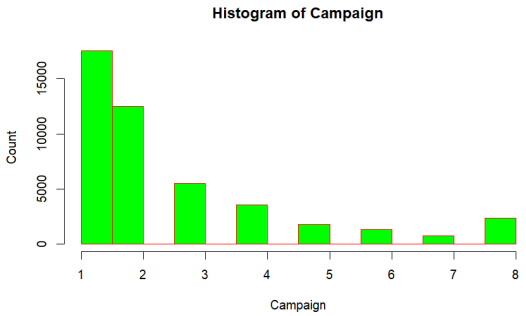

(b) Image 10



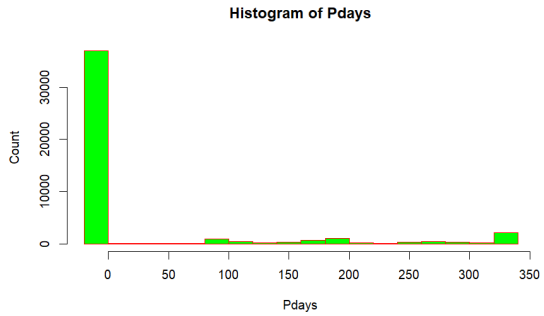

(c) Image 11



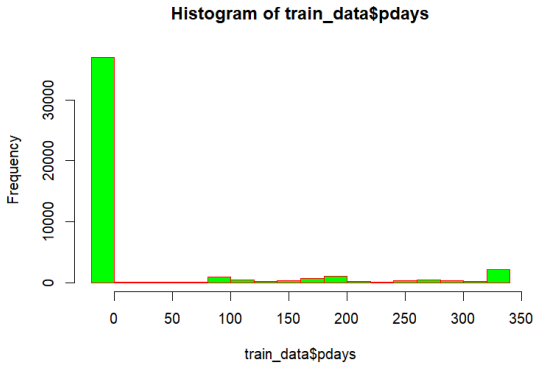

(d) Image 12



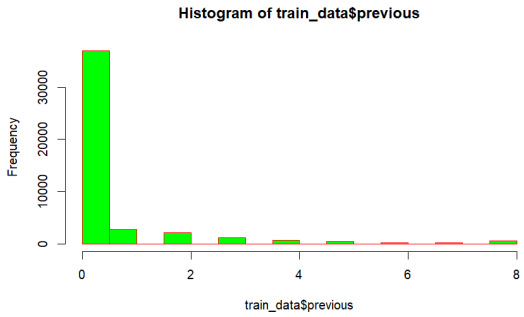

(e) Image 13




(f) Image 14



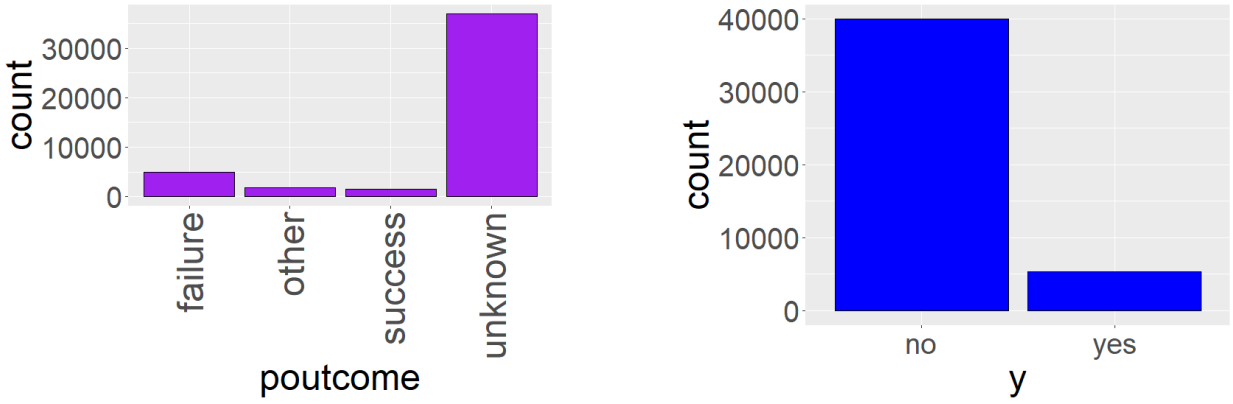

(g) Image 15




(h) Image 16



Figure 3: Multiple Histograms,Barplots arranged on this page




(a) Image 17

(b) Image 18

Figure 4: Multiple Histogram,Barplots arranged on one page

# 7 Encoded categorical Variables

In the dataset , several variables were originally categorical in nature, such as `job`, `marital`, `education`, `default`, `housing`, `loan`, `contact`, `month`, `poutcome`, and the target variable `y`. These variables contained string-based values and could not be directly used in most machine learning algorithms. To address this, we applied **label encoding**, transforming each unique category into a corresponding integer.

## Encoding Process

One-Hot Encoding converts categorical variables into a set of binary (0/1) columns, where each category is represented as a separate feature. This method avoids introducing any ordinal relationship between categories.

- It is suitable for nominal (unordered) categorical variables.
- Prevents unintended ordinal relationships between categories.
- Ensures compatibility with machine learning algorithms requiring numerical input.

## Variables Encoded

The following categorical variables were encoded using One-Hot Encoding:

- **job:** Converted into multiple binary columns (e.g., job_admin, job_technician, etc.).
- **marital:** Converted into binary columns (e.g., marital_single, marital_married).
- **education:** Converted into binary columns (e.g., education_primary, education_secondary).
- **default:** Converted into binary columns (default_yes).
- **housing:** Converted into binary columns (housing_yes).
- **loan:** Converted into binary columns (loan_yes).
- **contact:** Converted into binary columns (contact_cellular, contact_telephone).
- **month:** Converted into binary columns (month_jan, month_feb, etc.).
- **poutcome:** Converted into binary columns (poutcome_success, poutcome_failure, etc.).

## Handling Dummy Variable Trap

To avoid the dummy variable trap (multicollinearity), one category from each categorical variable was dropped during encoding. This ensures that the encoded variables remain independent and do not introduce redundant information into the model.

## Example Encoding Tables

Below are sample One-Hot Encoding representations after dropping one category:

| **Marital Status** | **Single** | **Married** |
|---|---|---|
| Single | 1 | 0 |
| Married | 0 | 1 |
| Divorced | 0 | 0 |

Table 1: One-Hot Encoding for `marital` (Divorced dropped)

| **Education Level** | **Primary** | **Secondary** |
|---|---|---|
| Primary | 1 | 0 |
| Secondary | 0 | 1 |
| Tertiary | 0 | 0 |

Table 2: One-Hot Encoding for `education` (Tertiary dropped)

## Conclusion

All categorical variables in the dataset were transformed using One-Hot Encoding. To prevent multicollinearity, one category from each variable was dropped (drop-first approach). This transformation ensures that the dataset is suitable for machine learning models while maintaining interpretability and avoiding redundant features.

## Discretization of `pdays`

The variable `pdays` in the dataset is a numeric (continuous) variable that represents the number of days passed since a client was last contacted in a previous campaign. A value of `-1` indicates that the client was not contacted before.

Working directly with this continuous variable may not capture the underlying patterns effectively. Therefore, we applied a **discretization strategy** to convert it into a binary categorical feature.

## Binning Strategy

We created a new feature based on the following rule:

- If $\texttt{pdays} = -1$, the client was **never contacted** before. We assign the new value as 0.
- If $\texttt{pdays} \geq 0$, the client was contacted previously. We assign the new value as 1.

$$\texttt{pdays\_bin} = \begin{cases} 0 & \text{if } \texttt{pdays} = -1 \\ 1 & \text{otherwise} \end{cases}$$

## Motivation and Benefits

- **Pattern Detection:** The transformation helps distinguish between clients who were previously contacted and those who were not.
- **Feature Simplification:** Reduces the dimensionality and variability of the data by converting a continuous variable into a simpler, more interpretable form.
- **Improved Model Performance:** Binary features can be more informative in classification tasks, especially for algorithms that benefit from categorical splits (e.g., decision trees).

By discretizing the `pdays` variable into a binary feature, we introduced a more informative attribute that captures client contact history in a meaningful way. This preprocessing step is aimed at enhancing model interpretability and predictive power.

# 8 Handle Imbalanced Data

In many real-world classification problems, the distribution of classes is often imbalanced. That is, one class (called the majority class) significantly outnumbers the other class(es) (called the minority class). Examples include fraud detection, medical diagnosis, and churn prediction.

Let us consider a binary classification task where:

- Class 0 (majority class): 88% of the data
- Class 1 (minority class): 12% of the data

If we train a model directly on this imbalanced data, it might predict class 0 for all instances and still achieve 90% accuracy, which is misleading and unhelpful.

## Why Transformation is Necessary

- **Bias in Model Training:** The model learns to favor the majority class, ignoring the minority class.
- **Poor Minority Class Performance:** Metrics such as recall, precision, and F1-score for the minority class are often very low.
- **Real-World Cost:** Misclassifying the minority class (e.g., failing to detect fraud) can be very costly.

There are several techniques to transform imbalanced data such as **random undersampling**, **random oversampling**, **SMOTE (Synthetic Minority Over-sampling Technique)**, **Ensemble Method** and **cost-sensitive learning**, which help balance class distribution and improve model performance.
In this dataset, class 0 represents 88% of the observations while class 1 represents only 12%, indicating a significant class imbalance. To address this, we apply **SMOTE(Synthetic Minority Over-Sampling Technique)** ,where we address class imbalance by generating synthetic samples for the minority class (class 1). Instead of randomly duplicating existing observations, SMOTE creates new samples by interpolating between neighboring minority class instances. This helps to balance the class distribution and reduces the risk of overfitting, while preserving information from the majority class.

# 9 Splitting Balanced Data into Training and Testing Sets

After performing data balancing (such as undersampling), the next critical step in the modeling process is to split the dataset into **training** and **testing** sets. In this context, we use an 80:20 split, where:

- **80% of the data** is used for training the model.
- **20% of the data** is held out for testing (validation) purposes.

### Why Perform the Split After Balancing?

Balancing is applied first so that the class distribution is even before the data is divided. If the split were done before balancing, the class proportions in the training and testing sets might remain imbalanced, defeating the purpose of balancing. By balancing first, both the training and testing datasets benefit from a fair class representation.

### Purpose of the Split

- The **training set** is used to allow the machine learning model to learn patterns and relationships within the data.
- The **testing set** is used to evaluate the model's performance on unseen data, providing an estimate of how well it will generalize to new cases.

Splitting the dataset into training and testing sets after balancing is a best practice in machine learning. It ensures that the model is trained on balanced data and evaluated fairly on a separate, balanced test set. This step plays a crucial role in building robust, generalizable models.

## 10 Feature Scaling

Feature scaling was applied to continuous variables to bring them into a common range, ensuring that no single feature dominates the model due to differences in magnitude. This is particularly important for machine learning algorithms that are sensitive to the scale of input features.

- Scaling helps in faster convergence of optimization algorithms.
- Prevents features with large values from dominating smaller ones.
- Improves overall model performance and stability.

## Method Used

Standardization (Z-score scaling) was applied to the continuous variables. This technique transforms the data such that it has a mean of 0 and a standard deviation of 1.

$$Z = \frac{X - \mu}{\sigma}$$

where:

- $X$ is the original value,
- $\mu$ is the mean of the feature,
- $\sigma$ is the standard deviation of the feature.

## Variables Scaled

The following continuous variables were scaled:

- **age**
- **balance**
- **duration**
- **campaign**
- **previous**

## Conclusion

Scaling of continuous variables ensured that all features contributed equally to the model, preventing bias due to differing scales. This preprocessing step improved the efficiency and predictive performance of the machine learning models.

# 11 Model Training, Prediction, and Performance Evaluation

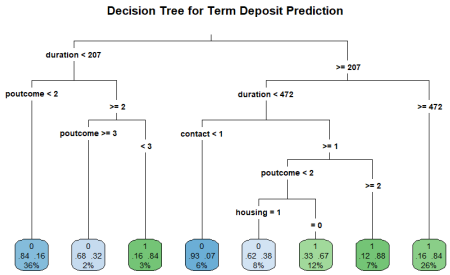
After preprocessing the Kaggle Bank Marketing dataset (including encoding categorical variables, discretizing `pdays`, handling outliers via capping, and balancing via SMOTE) and splitting it into training (80%) and testing (20%) sets, the following steps are performed to train models, make predictions, and evaluate performance.

## 11.1 Model Training

Four machine learning models are trained on the balanced training set $\mathcal{D}_{\text{train}}$ to predict the binary target variable $y$ (0 = no subscription, 1 = subscription to term deposit):

- **Decision Tree:** A classifier that recursively splits the feature space based on feature values to maximize information gain. Hyperparameters, such as maximum depth and minimum samples per split, are tuned using cross-validation.Optimized model performance using hyperparameter tuning, reducing overfitting and improving generalization on unseen data.

  The plot of the desicion tree is -




The most important variable is "Duration" ,from this variable we start the splitting.Therotically we use Entropy,Information Gain and Gini Index metrics to find the most important variable.

- **Random Forest:** An ensemble of decision trees that aggregates predictions via majority voting to enhance robustness and reduce overfitting. Key hyperparameters include the number of trees and maximum depth, tuned via cross-validation.Optimized model performance using hyperparameter tuning, reducing overfitting and improving generalization on unseen data. Random Forests are ensembles of decision trees, and while we can't directly "plot" all trees at once, you can visualize a few individual trees and get an idea of how they combine.From out-of-Bag error plot we fix the number of trees at 40. Just seeing a desicion tree-

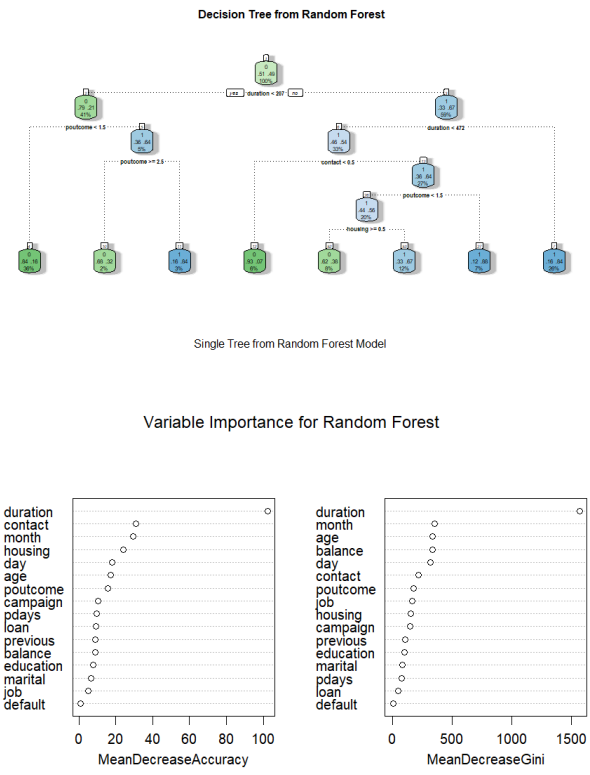



Figure 5: Caption

The most important variable is "Duration".From here the splits start.We know it from a diagram

- **Support Vector Machine (SVM):** A classifier that finds the optimal hyperplane to separate classes, using a kernel (e.g., RBF) for non-linear data. The regularization parameter $C$ and kernel parameters (e.g., $\gamma$ for RBF) are tuned to balance margin maximization and error.
- **XGBoost:**An optimized gradient boosting algorithm that builds an ensemble of decision trees sequentially, where each tree aims to correct the errors of the previous ones. It incorporates regularization to control model complexity and prevent overfitting. Key hyperparameters, including the number of estimators, learning rate, and maximum depth, are tuned using cross-validation to enhance predictive performance..

Each model is trained to minimize its respective loss function on $\mathcal{D}_{\text{train}}$, leveraging the balanced class distribution achieved through undersampling.

## 11.2 Prediction

The trained models are used to predict the target variable $y$ on the test set $\mathcal{D}_{\text{test}}$:

- Generate class predictions ($\hat{y} \in \{0, 1\}$) for each instance in $\mathcal{D}_{\text{test}}$ using the trained models.

## 11.3 Performance Evaluation

Model performance is evaluated on $\mathcal{D}_{\text{test}}$ using the confusion matrix and standard metrics:

- **Confusion Matrix:** A table summarizing prediction outcomes:

| | Predicted: 1 | Predicted: 0 |
|---|---|---|
| Actual: 1 | TP | FN |
| Actual: 0 | FP | TN |

where:

  - TP: True Positives (correctly predicted subscriptions).
  - TN: True Negatives (correctly predicted non-subscriptions).
  - FP: False Positives (incorrectly predicted subscriptions).
  - FN: False Negatives (missed subscriptions).
  - **Accuracy:** Proportion of correct predictions:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$

    where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively.
  - **Precision:** Proportion of positive predictions that are correct:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

  - **Recall:** Proportion of actual positives correctly identified:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

  - **F1-Score:** Harmonic mean of precision and recall:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

The models are compared based on these metrics to identify the best-performing classifier for predicting term deposit subscriptions. Results are summarized in a table: The values in the table are computed

Table 3: Performance Metrics for Classification Models

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Decision Tree | 0.9353 | 0.9337 | 0.93745 | 0.9355 |
| Random Forest | 0.9508 | 0.9604 | 0.9405 | 0.9503 |
| SVM | 0.9373 | 0.9310 | 0.9448 | 0.9379 |
| XGBoost | 0.9463 | 0.9547 | 0.9459 | 0.9373 |

post-evaluation and reflect model performance on $\mathcal{D}_{\text{test}}$.

# Model Comparison and Best Model Selection

In above table presents the performance metrics—Accuracy, Precision, Recall, F1-Score—for four machine learning models: Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost. These metrics are crucial in evaluating how well each model predicts whether a client will subscribe to a term deposit.Here for imbalance dataset F1-Score is important to check which one is better.

## Model-Wise Interpretation

- **Decision Tree:** The Decision Tree model achieves an accuracy of 93.53% with balanced precision (93.37%) and recall (93.75%). While its performance is good, it is slightly lower compared to ensemble methods. However, it remains highly interpretable and useful in scenarios where model transparency is important.
- **Random Forest:** The Random Forest model demonstrates excellent performance with the highest accuracy (95.08%) and precision (96.04%). It also maintains strong recall (94.05%) and F1-score (95.03%), making it a highly reliable and robust model for classification tasks.
- **Support Vector Machine (SVM):** SVM achieves an accuracy of 93.73% with a good balance between precision (93.10%) and recall (94.48%). Its F1-score (93.79%) indicates stable performance, making it a competitive model for binary classification problems.
- **XGBoost:** XGBoost delivers strong performance with an accuracy of 94.63%, precision of 95.47%, and recall of 94.59%. Its ability to capture complex patterns and control overfitting makes it one of the most powerful ensemble methods, although its F1-score (93.73%) is slightly lower than Random Forest.

## Conclusion and Best Model

Based on the performance metrics:

- **Best Overall Model: Random Forest** is the top-performing model, achieving the highest accuracy, precision, and F1-score, making it the most balanced and reliable model.
- **Alternative Choice: XGBoost** is a strong alternative, offering excellent performance and better handling of complex relationships in the data.
- **Use Cases for SVM and Decision Tree:** SVM provides stable and balanced performance, while Decision Tree is most suitable when interpretability and simplicity are prioritized over slightly higher accuracy.

Thus, **Random Forest** is recommended as the best model for predicting term deposit subscriptions in this study.